

# MD RAKIBUL ISLAM RAIHAN

## AI Engineer - LLM Inference & Multimodal Research (2027 New Grad)

Xi'an, China | Open to Relocation | Available July 2027

Nationality: Bangladesh | Sex: Male | DOB: 2002.12.06

Languages: Bengali (Native) | English (Fluent) | Chinese (Conversational)

Work Authorization: Requires China work visa (employer-sponsored Z Visa) | Global: Sponsorship welcome

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Portfolio: theraihanrakibb.github.io/Online-Portfolio



## SUMMARY

2027 New Grad AI Engineer focused on LLM inference systems and multimodal research. Built and deployed LLM serving on **4x A800** clusters at **InfIX.ai** (SGLang, KV-cache offload, PD-disaggregation - **92.27% cache hit, 8.3x faster TTFT**). **First author**, IEEE TNNLS (Q1). Seeking AI / Research Engineer roles in China, Hong Kong, and globally.

## EDUCATION

**M.Eng. in Software Engineering**, School of Software, Northwestern Polytechnical University (985/211) - Sep 2024 - Jul 2027

GPA: 88/100 | Top 1% | Chinese Government Scholarship | President of International Student Union (ISU)

**B.Eng. in Computer Science & Technology**, School of Computer Science, Northwestern Polytechnical University (985/211) - Sep 2020 - Jul 2024

GPA: 85/100 | Top 1% | Presidential Scholarship | Best Undergraduate Thesis Award

## EXPERIENCE

**AI Infrastructure Engineer Intern | InfIX.ai | Shenzhen, China | Apr 2026 - Jun 2026**

- Cut TTFT 8.3x (55s -> 6.7s) by building a 3-tier KV-cache offload (GPU HBM -> CPU DRAM -> SSD) over Mooncake RDMA on 4x A800, reaching 92.27% cache-hit for long-context models.
- Patched SGLang source across cache-management and memory-allocation modules (kvcacheio.py, hiradix\_cache.py, memory\_pool\_host.py) to fix production cache failures and deploy hybrid-attention Qwen models.
- Built PD-disaggregated (prefill/decode) serving with RDMA KV sharing, raising stable concurrency 192 -> 256 req and boosting GPU utilization via RadixAttention prefix-caching and continuous batching.
- Profiled CUDA/Triton inference with Nsight Systems, localized GEMM bottlenecks, and improved serving throughput via kernel optimization and FP8 quantization.
- Applied Little's Law to model capacity limits, proving long-context serving on 4x A800-80GB was bounded to ~1.5 RPS and guiding scale-out.
- Diagnosed and resolved 12+ production incidents across RDMA, etcd, networking, storage, and distributed inference.
- Deployed Qwen models on A800 clusters (TP2 single-node, TP4 with PD-disaggregation) and ran DeepSeek on H800 multi-node clusters (8 GPUs each), tuning tensor-parallel layouts for throughput and stability.

**Software Engineer Intern (AI Agent) | Hong Kong Amram Group | Shenzhen, China | Jun 2025 - Sep 2025**

- Developed production full-stack applications using React, REST APIs, and backend automation services.
- Built AI-agent workflow automation systems that improved operational efficiency by **25%**.

**Electrical Software Engineer Intern | Shaanxi Longong Intelligent Technology | Xi'an, China | Feb 2025 - Apr 2025**

- Developed PLC-based industrial control systems using CODESYS for robotics and industrial automation.
- Implemented Modbus, CAN, and Ethernet/IP communication protocols, improving motor-control performance by **15%**.

## RESEARCH

**Graduate Researcher | Northwestern Polytechnical University | Sep 2024 - Present**

Advisor: Prof. Zhang Peng

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer** achieving **4x memory reduction** while maintaining generation quality.
- Implemented low-precision quantization (**FP8/FP4/NVFP4**) for multimodal diffusion transformers; evaluated quality-efficiency trade-offs.
- Benchmarked NVFP4-DiT on **A800/H100 GPUs**; profiled end-to-end latency and memory with PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

**Undergraduate Researcher | Northwestern Polytechnical University | Jan 2023 - Jul 2024**

Advisor: Dr. Yaxing Chen

- Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, achieving **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

## PROJECTS

**AI Infra Portfolio | github.com/theraihanrakibb/ai-infra-portfolio**

- 10 production-grade, Dockerized AI-infra projects: LLM gateway, model serving, GPU observability, training ops, MLOps registry, KV-cache simulator, benchmarking, quantization, tracing, RAG.

**Distributed Training Launcher | github.com/theraihanrakibb/train-launcher**

- CLI for multi-node PyTorch training with fault tolerance and automatic retry; cut manual ops overhead for distributed jobs.

**C Compiler (Frontend) | github.com/theraihanrakibb/c-compiler-frontend**

- Self-built C compiler (lexer/parser/codegen) using Yacc/Lex; applied systems and algorithms fundamentals.

**NVFP4-DiT (Multimodal Generation) | github.com/theraihanrakibb/NVFP4-DiT**

- First-author IEEE TNNLS (Q1, under review) 4-bit audio-guided video diffusion transformer; 4x memory reduction while preserving generation quality on A800/H100.

**Multimodal Deepfake Detection | github.com/theraihanrakibb/M.Eng-Thesis-Multimodal-Deepfake-Audio-Visual-Temporal-Framework**

- M.Eng. thesis: audio-visual-temporal framework detecting deepfake video via visual, audio, and temporal inconsistency analysis.

## TECHNICAL SKILLS

**Languages:** Python (primary) | C | C++ | Java | SQL | Bash

**AI / ML:** PyTorch | TensorFlow | Transformers | scikit-learn | NumPy | deep learning | model evaluation & metrics

**LLMs & serving:** LLM inference & serving (SGLang, vLLM) | RAG | agents | fine-tuning (LoRA/QLoRA) | quantization (FP8/INT4)

**GPU & systems:** CUDA | Triton | FlashAttention | multi-GPU/distributed training (DDP, FSDP) | PyTorch Profiler | Nsight

**Computer vision & generative AI:** CNNs | Vision Transformers | diffusion models (DDPM/DiT) | video generation | multimodal learning | OpenCV

**Data & MLOps:** Pandas | Docker | Kubernetes | Git | CI/CD | Slurm | AWS / GCP | Apache Spark | Weights & Biases / MLflow

**Math & theory:** linear algebra | probability & statistics | optimization | convex optimization | information theory

**Fundamentals:** data structures & algorithms | concurrency | operating systems | computer networks | distributed systems

## PUBLICATIONS & HONORS

**Publication:** First Author, "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" - IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), Under Review.

**Honors:** Chinese Government Scholarship (2024-2027); Presidential Scholarship (2020-2024); Best Undergraduate Thesis Award (2024); Outstanding Graduate (2024), NWPU; Hackathon Winner - NWPU AI Challenge (2023).